

From “Zero Worlds” to Quantum Worlds?

Observer-First Foundations and the Reconstruction of Quantum Mechanics

Executive summary

The strongest interpretation of a “Zero Worlds” research program is not merely that spacetime is emergent, or that quantum states are epistemic, but that **no observer-independent physical world is fundamental at all**. Instead, one begins with first-person states, experiences, informational patterns, or abstract computational relations and asks whether the familiar physical world—including quantum theory—can be reconstructed rather than assumed. Your uploaded essay uses “zero worlds” in precisely this sense and, importantly, identifies the relevant Müller as **Markus P. Müller**, author of *Law without Law* and the 2026 *Algorithmic Idealism* paper. The uploaded essay also correctly distinguishes its own speculative synthesis involving synchronicity from Müller’s actual claims. ¹ Müller’s primary work formulates the fundamental law in terms of observer/self-states and universal induction, while treating an apparently objective external world as an emergent statistical regularity.

The central finding of this review is that **no presently developed consciousness-first, observer-first, or ruliad-first theory has rigorously derived the full mathematical and empirical content of quantum mechanics from its distinctive non-material primitives**. In particular, none of Müller’s Algorithmic Idealism, Hoffman’s Conscious Agent Theory, or Wolfram’s ruliad program currently derives, from its deepest postulates alone, the complete package

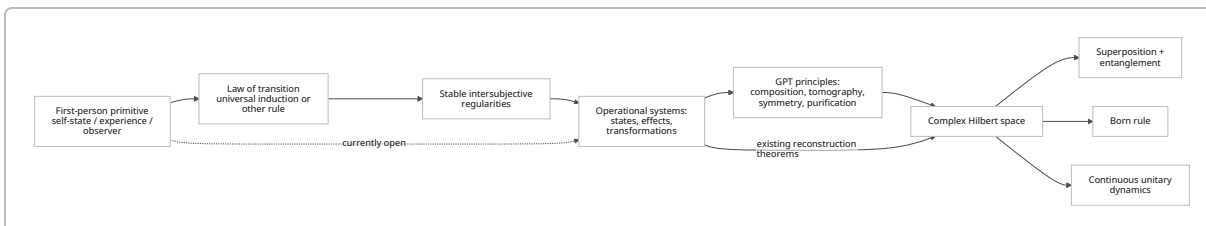
{complex Hilbert space, tensor-product composition, superposition, entanglement, unitary dynamics, Born prob

Müller himself is unusually explicit about this limitation: in 2026 he identifies Bell-inequality violation together with no-signalling and efficient BQP quantum computation as future “litmus tests” for Algorithmic Idealism, and says that merely obtaining mathematical objects resembling wavefunctions or complex numbers would not suffice. ² Wolfram argues that quantum mechanics follows generically for computationally bounded, persistent observers embedded in the ruliad, but his own 2026 presentation acknowledges that all details of those derivations are not yet known. ³ Hoffman and Prakash obtain rigorous Markov-chain constructions and a formal correspondence between certain harmonic functions of conscious-agent dynamics and a free-particle wavefunction, but that is far short of a reconstruction of general quantum theory. ⁴

There **is**, however, a rigorous route from comparatively ontology-light assumptions to quantum mechanics: the operational and information-theoretic reconstruction literature. Hardy, Chiribella–D’Ariano–Perinotti, and Masanes–Müller show that suitably chosen axioms concerning information capacity, reversible transformations, composition, distinguishability, compression, causality, and purification can recover finite-dimensional complex quantum theory without taking Hilbert space itself as a primitive. ⁵ These reconstructions are mathematically much stronger than the quantum derivations presently available in

Zero-Worlds-style programs—but they are **not themselves idealist derivations**. They start with operational systems, preparations, transformations, measurements, and probabilities. The key research opportunity is therefore to connect the two levels.

The most promising architecture for a serious Zero Worlds program is consequently a **two-stage reconstruction**:



The **right-hand half** of this diagram is already supported by rigorous reconstruction theorems. The **left-hand bridge**—showing why an observer-first law necessarily produces precisely those operational principles rather than classical probability theory, real/quaternionic quantum theory, a generic generalized probabilistic theory, or a supra-quantum no-signalling theory—is the missing theorem. ⁶

Empirically, existing Bell experiments, contextuality experiments, interference, quantum computing, and extended Wigner-friend results strongly constrain candidate reconstructions, but **they do not presently discriminate in favor of idealism, consciousness-first ontology, the ruliad, or Algorithmic Idealism over empirically adequate physicalist interpretations of quantum mechanics**. Loophole-free Bell experiments reject broad classes of local-realist models while agreeing with quantum theory; they do not establish that matter is emergent from observers. ⁷ Extended Wigner-friend/“local friendliness” results likewise place pressure on packages of assumptions about observer-independent facts and locality, but they do not uniquely select a Zero Worlds ontology. ⁸

The most defensible claim for a publication site, therefore, is stronger than “this is all speculation” but substantially weaker than “quantum mechanics has been derived from consciousness.” It is:

There is now a mathematically serious family of observer-first, information-first, and computation-first research programs in which objective physical reality is demoted from primitive status. Rigorous reconstruction theory demonstrates that much of finite-dimensional quantum mechanics can be recovered from abstract operational/informational principles. What has not yet been shown is that the distinctive postulates of the strongest Zero Worlds theories entail those principles, or uniquely entail quantum rather than other probabilistic physics.

That gap is sufficiently precise that it can be turned into a concrete mathematical research agenda.

What “Zero Worlds” means and where the major programs sit

“Non-physical” can mean several very different things, and collapsing them together makes the literature look more unified than it is. At least four levels should be distinguished.

Strong Zero Worlds idealism takes first-person states or consciousness itself as fundamental and denies that an observer-independent physical world is ontologically primitive. Markus P. Müller’s Algorithmic Idealism is the cleanest current formal example: its fundamental objects are abstract self-states and probabilistic transitions determined by a universal method of induction. Müller emphasizes that concepts such as “agent” and “consciousness” need not be primitive; self-states can simply be abstract patterns. ⁹ Hoffman’s conscious realism is stronger phenomenologically but less deflationary: conscious subjects and experiences are fundamental, and the “world” is ultimately constituted by conscious agents rather than spacetime objects. ⁴

Computational or mathematical monism makes an abstract structure fundamental rather than experience. Wolfram’s ruliad belongs here. The ruliad is supposed to be the entangled limit of all possible computations, with physical reality as perceived by us arising from the particular way computationally bounded, temporally persistent observers sample or coarse-grain it. Wolfram therefore does not advocate “zero ontology” in Müller’s sense: there is a large observer-independent abstract structure underneath experience. ³

Agent-centered or operational quantum foundations—including QBism and generalized-probabilistic-theory reconstructions—demote the quantum state from a straightforward physical object but do not generally abolish an external world. QBism regards a quantum state as an agent’s probability assignment and interprets the Born rule as an additional normative consistency relation connecting an agent’s probabilities. It nevertheless retains systems and their objective capacities as part of its conceptual vocabulary. ¹⁰ Operational reconstructions similarly remain neutral about deep ontology while assuming experimentally meaningful preparations, transformations, and outcomes. ¹¹

Transformation-first and epistemically restricted physical theories—constructor theory and Spekkens-style epistemic models—are valuable comparisons but are not Zero Worlds theories. Deutsch explicitly describes constructor theory in physical terms: information is instantiated in physical substrates and the fundamental statements concern which physical transformations are possible or impossible. ¹² Spekkens’ toy theory assumes underlying ontic states but restricts an observer’s possible knowledge of them. ¹³

Program	Fundamental starting point	Is a physical world fundamental?	Main mathematics	What it presently establishes about QM	Best classification
M. P. Müller: Algorithmic Idealism	Self-states + universal induction	No	Algorithmic information theory, Solomonoff induction	Emergence of simple computable probabilistic “worlds”; quantum nonclassicality remains an explicit research target. ¹⁴	Strong Zero Worlds

Program	Fundamental starting point	Is a physical world fundamental?	Main mathematics	What it presently establishes about QM	Best classification
Hoffman / Prakash	Conscious experiences and conscious agents	No, not spacetime physics	Measurable spaces, Markov kernels/ chains, combinatorics	Free-particle-wavefunction-like correspondence; no full Hilbert/ Born reconstruction. ⁴	Consciousness-first idealism
Wolfram Physics Project	Ruliad/all possible computations	Not material, but yes as abstract structure	Hypergraph rewriting, multiway systems, causal/ branchial graphs	Claimed generic emergence of QM for bounded observers; formal correspondences exist, but complete derivation is unfinished. ¹⁵	Computational monism
QBism	Agent's personal probabilities/ actions/ experiences	External systems remain	Bayesian probability, SIC representation	Exact reformulation of QM; Born rule treated as empirical normative addition, not derived from pure Bayesianism. ¹⁰	Agent-centered interpretation
Hardy / CDP / Masanes-Müller	Operational information principles	Ontologically neutral	Convex geometry, GPTs, group theory	Rigorous reconstructions of finite-dimensional QM under explicit axioms. ⁵	Operational reconstruction
Constructor theory	Possible/ impossible physical tasks	Yes	Task algebra, substrates, attributes	Recovers qualitative quantum-information phenomena in "superinformation" theories, not a unique full derivation of complex QM. ¹⁶	Transformation-first physicalism

Program	Fundamental starting point	Is a physical world fundamental?	Main mathematics	What it presently establishes about QM	Best classification
Spekkens epistemic program	Ontic states + epistemic restriction	Yes	Discrete probability/hidden variables	Reproduces many quantum analogues but explicitly fails Bell and Kochen–Specker phenomena. ¹³	ψ -epistemic model
ψ-ontology no-go results	General ontological models	Usually yes	Probability over hidden ontic states	PBR excludes broad overlapping ψ -epistemic models given preparation independence. ¹⁷	Constraint on ontic/epistemic models

The Müller ambiguity. For the specific Zero Worlds context, the intended person is almost certainly **Markus P. Müller**. Your uploaded document explicitly names him on page 2 and bases its “zero-worlds” discussion on *Law without Law* and Algorithmic Idealism. ¹⁸ Markus Müller is doubly relevant because, independently of Algorithmic Idealism, he coauthored with Lluís Masanes one of the strongest operational derivations of finite-dimensional quantum theory. ¹⁸

A historically relevant **Johannes Peter Müller** should nevertheless be mentioned as a possible source of the name confusion. The nineteenth-century physiologist formulated the doctrine or law of **specific nerve energies**: the qualitative character of sensation depends on the sensory pathway activated rather than simply mirroring the external stimulus. That makes him an important historical predecessor of the idea that perception is interface-dependent, especially in the lineage leading through Helmholtz, but he is **not a modern quantum-foundations theorist and did not propose a Zero Worlds derivation of physics**. ¹⁹

Donald Hoffman is particularly interesting in this historical context. The Fitness-Beats-Truth theorem of Prakash, Stephens, Hoffman, Singh, and Fields analyzes evolutionary strategies under specified game-theoretic assumptions and finds conditions under which fitness-only perceptual strategies outperform truth-tracking ones. ²⁰ This supports an argument for perception-as-interface, but the logical chain must be kept straight:

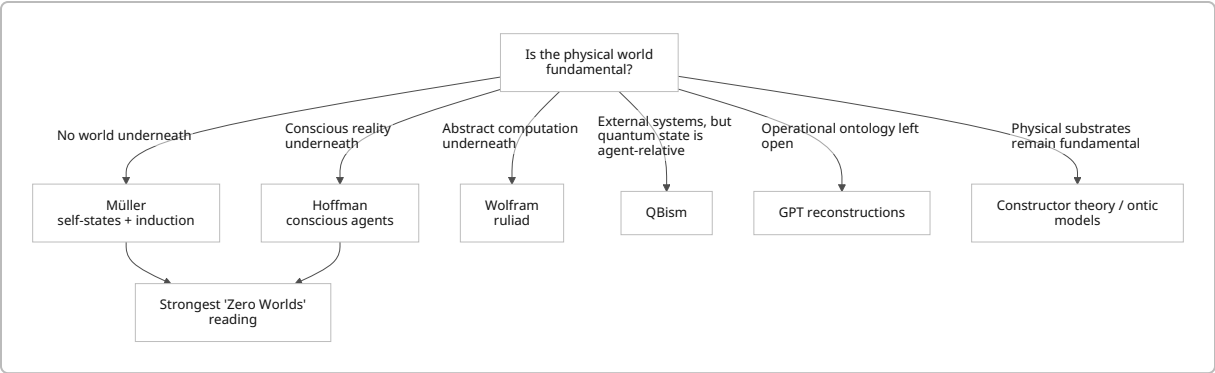
selection need not optimize veridical perception $\nRightarrow$ consciousness is ontologically fundamental $\nRightarrow$ quantum mechan

The first arrow is an additional metaphysical hypothesis; the second is a major mathematical research problem. Hoffman and Prakash themselves present conscious realism as a hypothesis to be developed and tested, not as something already proved by evolutionary theory. ²¹

Wolfram’s position is similarly easy to overstate. His strongest claim is not merely that cellular automata can imitate quantum systems. He argues that once all possible computation is represented in the ruliad, an observer with limited computational power and a persistent identity necessarily coarse-grains that

underlying structure in ways that generate the major laws associated with relativity, quantum mechanics, and thermodynamics. ³ But Wolfram also explicitly discusses unresolved circularity in defining observers and states that further observer characteristics will be needed to derive more detailed physics, such as specific particle-physics structure and parameters. ²²

The intellectual family resemblance is therefore real, but the ontologies differ radically:



Starting postulates: what is actually being assumed

A serious reconstruction must keep an **assumption ledger**. “Starting from nothing” is never literally true: even the most austere program assumes mathematical objects, rules of inference, probability structure, or constraints on observers. The scientific question is whether those assumptions are more primitive, independently motivated, and less tailored to quantum mechanics than the Hilbert-space postulates they are supposed to explain.

Assumption ledger

Approach	Primitive postulates or assumptions	What is <i>not</i> initially assumed	Hidden cost / major unresolved issue
Algorithmic Idealism	Self States: there is a set $\mathcal{S}$ of possible first-person informational patterns. State Change: the objective likelihood of the next self-state is supplied by a universal method of induction. In the bit model this is based on algorithmic probability.	Fundamental spacetime, matter, global world-state, fixed physical laws	Choice/robustness of formal self-state representation; universal-machine dependence; handling forgetting; derivation of specifically quantum statistics.

Approach	Primitive postulates or assumptions	What is <i>not</i> initially assumed	Hidden cost / major unresolved issue
Hoffman conscious realism	Conscious experiences and actions; perceptual, decision, and action channels represented by Markov kernels; ultimately the “world” consists entirely of conscious agents. ²¹	Fundamental spacetime and unconscious material objects	Why these particular Markov networks yield our exact physics; relation of mathematical agent to phenomenal consciousness; Born rule and QFT absent.
Wolfram/ruliad	All possible computations/rules; computational irreducibility; observer computational boundedness; observer persistence/equivalencing. ³	A single fundamental spacetime geometry or selected microscopic physical rule	Ruliad is already an enormous ontological/mathematical structure; observer characterization involves admitted circularities; exact QM measure remains underdeveloped.
QBism	Personalist Bayesian probabilities; measurements as agent actions; outcomes as agent experiences; quantum dimension/capacity associated with a system; Born rule as empirical normative constraint. ¹⁰	Ontic wavefunction	Does not attempt to eliminate an external world or derive all quantum structure from pure probability theory.
Chiribella-D’Ariano-Perinotti	Causality, perfect distinguishability, ideal compression, local distinguishability, pure conditioning, plus purification . ²³	Hilbert space as a primitive	Why nature satisfies these information principles remains an explanatory question.
Masanes-Müller	Finite information/state description, local tomography/composition, equivalence principles for systems/subspaces, reversible symmetry, and sufficiently rich measurements. ¹⁸	Complex vector spaces and trace rule as primitives	Operational assumptions are strong; connecting them to observer-first ontology is open.
Hardy	Five operational axioms, including continuous reversible transformations between pure states. ²⁴	Standard Hilbert postulates	Continuity is the crucial selector between classical and quantum possibilities; its deeper origin is unexplained.

Approach	Primitive postulates or assumptions	What is <i>not</i> initially assumed	Hidden cost / major unresolved issue
Constructor theory	Fundamental statements specify possible and impossible transformations/tasks; information is physically instantiated. ¹²	Initial-condition-plus-equation-of-motion framework as fundamental explanatory language	Not ontology-free: it explicitly regards information as physical.
Spekkens toy theory	Underlying ontic states plus a knowledge-balance restriction: maximal knowledge leaves as many fundamental questions unanswered as answered. ¹³	Quantum amplitudes	Its local epistemic restriction cannot reproduce Bell or Kochen–Specker contextuality.

For Algorithmic Idealism, the formal core can be made particularly explicit. In the bit-model idealization, let an observer’s current informational state be a finite bit string x . A universal monotone machine U defines an algorithmic semimeasure of the schematic form

$$M_U(x) = \sum_{p: U(p)=x*} 2^{-|p|},$$

where the sum runs over programs whose output begins with x . Conditional continuation weights are obtained from ratios such as

$$M_U(y \mid x) \sim \frac{M_U(xy)}{M_U(x)},$$

with normalization subtleties because Solomonoff’s object is generally a semimeasure rather than an ordinary computable probability distribution. At heuristic coding-theorem level,

$$M_U(y \mid x) \approx 2^{-K_U(y|x)}$$

up to multiplicative constants and formulation-dependent qualifications. Thus continuations that are algorithmically simple given the current state receive larger weight. Müller’s 2026 presentation deliberately allows Postulate 2 to have more than one mathematical realization rather than identifying Algorithmic Idealism with one fixed bit model. ²⁵

The rigorous achievement is not merely that “simple things are likely.” Solomonoff induction has convergence properties. If observations are in fact generated by a computable measure μ , then under the standard conditions Müller invokes,

$$\lim_{n \rightarrow \infty} |M(b \mid x_1 \dots x_n) - \mu(b \mid x_1 \dots x_n)| = 0$$

for the next bit b , with μ -probability one. Müller uses this to argue that observers with sufficiently information-rich self-states will experience themselves as embedded in stable, simple, probabilistic worlds governed by computable laws. ²⁶

That result is substantial, but its exact explanatory reach is easy to misunderstand. It gives:

universal induction + data generated by computable law $\mu \implies$ predictions converge to μ .

It does **not** by itself give

$$\text{universal induction} \implies \mu = \mu_{\text{quantum}}.$$

This distinction is the central technical issue for a Zero Worlds derivation of quantum mechanics.

Hoffman's primitive is different. A conscious agent is represented, suppressing measure-theoretic details, by something of the form

$$C = (X, G, P, D, A, N),$$

where X is an experience space, G an action space, P , D , and A are Markovian perceptual, decision, and action kernels, and N indexes updates. Hoffman and Prakash then impose **conscious realism**: ultimately, what plays the role of the agent's world consists of other conscious agents. Networks of agents therefore induce Markov dynamics. ²¹ The formalism is mathematically ordinary probability theory at the fundamental layer; the radical move is ontological, namely interpreting its primitive states as consciousness rather than physical configurations.

Wolfram makes almost the opposite move: rather than a tiny ontology, he begins with an extremely large mathematical one—the totality of computation. A rule such as

$$H \longrightarrow H'$$

rewrites a discrete combinatorial structure. Allow every admissible rewrite, and one obtains a **multiway system**; identify causal relations between events, and one obtains causal graphs; compare different branches at a given stage, and one obtains “branchial” structure. The ruliad is then the universal limit encompassing all such computations. Observers coarse-grain vast numbers of microscopic computational distinctions. ³

This yields an important philosophical caution: **non-material is not the same as assumption-free**. Müller assumes a universal induction principle; Hoffman assumes phenomenal primitives plus Markov structure; Wolfram assumes the ruliad and computational principles; informational reconstructions assume operational constraints. Their parsimony has to be assessed by explanatory output per assumption, not by simply counting the number of sentences called “postulates.”

Can quantum mechanics actually be recovered?

A useful rigor scale is:

Rigorous means a theorem whose conclusion is quantum theory given explicitly stated mathematical assumptions. **Conditional** means an exact consequence after part of quantum structure is already supplied. **Partial** means a correspondence with one structural component of quantum mechanics. **Speculative** means a proposed mechanism for which the required theorem has not yet been supplied.

On that scale, the strongest available route is not to try to jump directly from “observer” to “wavefunction.” It is to factor the derivation into stages.

Observer probabilities to an operational state space. Suppose an observer-first theory supplies probabilities

$$P(o \mid a, x)$$

for experiencing outcome o after performing an available intervention a , conditional on present self-state x . Define two self-states to be operationally equivalent when they yield identical future experimental probabilities:

$$x \sim x' \iff P(o \mid a, x) = P(o \mid a, x') \quad \forall a, o.$$

Each equivalence class then defines an **operational state** ω . An outcome o of intervention a defines an effect

$$e_{a,o}(\omega) = P(o \mid a, \omega).$$

If probabilistic mixtures of preparations are possible, then

$$\omega = \lambda\omega_1 + (1 - \lambda)\omega_2, \quad 0 \leq \lambda \leq 1,$$

and the state space is convex. Effects are affine maps into $[0, 1]$. Physical transformations become positive affine maps between state spaces. This is the basic generalized-probabilistic-theory architecture used by operational quantum reconstruction programs. ¹¹

This first step is mathematically straightforward **once the relevant conditional probabilities and intervention structure exist**. The difficult Zero Worlds question is why a theory of bare self-state transitions should furnish a sufficiently stable notion of intervention, repeatability, independent systems, composition, and intersubjective experiment to justify the GPT framework. That bridge is presently not derived by Müller’s core postulates. His 2026 paper explicitly contrasts Algorithmic Idealism with standard operational frameworks in which Alice and Bob are already assumed to be jointly embedded in a physical system. ²⁷

From GPT structure to complex quantum theory. This is where the existing reconstruction literature becomes powerful. Masanes and Müller show that a set of operational requirements concerning information capacity, tomography, equivalence of systems/subspaces, reversible symmetry and allowed measurements leads to the standard finite-dimensional quantum formalism. ¹⁸ Chiribella, D’Ariano and Perinotti instead use five general information-processing axioms plus the decisive principle of **purification**, deriving major quantum structures without initially postulating Hilbert space. ²³ Hardy’s earlier reconstruction similarly shows how continuous reversible transformation between pure states distinguishes quantum from classical probability theory within his axiomatic framework. ²⁴

Once this classification has produced a complex Hilbert space $\mathcal{H}$, the standard quantum ingredients no longer need independent metaphysical miracles.

The state space is

$$\Omega(\mathcal{H}) = \{\rho \in \text{Herm}(\mathcal{H}) : \rho \geq 0, \text{Tr}\rho = 1\}.$$

Pure states are rank-one projectors

$$\rho_\psi = |\psi\rangle\langle\psi|.$$

Because $\mathcal{H}$ is a complex vector space, for any $|0\rangle, |1\rangle$,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1,$$

is again a possible pure state. **Superposition has therefore been recovered once the complex linear state space has been derived.** This is rigorous conditional on the reconstruction axioms; it has not been independently derived from Algorithmic Idealism, Hoffman's consciousness ontology, or the ruliad. 28

For two systems,

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

The tensor product contains states not expressible as products. For example,

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

cannot be written as $|\psi_A\rangle \otimes |\psi_B\rangle$. **Entanglement therefore follows structurally from the quantum composition rule.** In purification-based reconstructions it takes on an even deeper role: a mixed state is represented as the marginal of an appropriate pure state of a larger system, with purification unique up to reversible transformation on the purifying subsystem. 23

Measurement effects become positive operators satisfying

$$0 \leq E \leq I,$$

and probabilities are

$$P(E | \rho) = \text{Tr}(\rho E).$$

Thus the **Born rule** is recovered in the operational reconstruction once the quantum state/effect cones and their pairing have been characterized. Again, this is a theorem under reconstruction axioms, not a derivation from idealism alone. 11

Continuous reversible evolution has the form

$$\rho(t) = U(t)\rho(0)U(t)^\dagger,$$

with

$$U(t) = e^{-iHt/\hbar}$$

for a continuous one-parameter unitary group. In these reconstruction programs the reversible transformation group is derived or constrained operationally; identifying its continuous time subgroup yields ordinary unitary quantum dynamics. ²⁸ What remains beyond such finite-dimensional reconstructions is the much harder task of deriving **which Hamiltonian**, spatial locality, gauge symmetry, relativistic QFT, the Standard Model representation content, and its numerical parameters.

QBism provides another useful exact calculation because it demonstrates what “Born rule as a probability relation” can mean without treating a wavefunction as a physical field. Suppose a symmetric informationally complete measurement exists in dimension d , with rank-one projectors Π_i and POVM elements

$$H_i = \frac{1}{d} \Pi_i.$$

Define

$$p(i) = \text{Tr}(\rho H_i) = \frac{1}{d} \text{Tr}(\rho \Pi_i).$$

Then the density matrix can be reconstructed from the probability vector as

$$\rho = \sum_{i=1}^{d^2} \left[(d+1)p(i) - \frac{1}{d} \right] \Pi_i.$$

For an actual measurement $\{D_j\}$, define

$$r(j | i) = \text{Tr}(D_j \Pi_i).$$

The Born probability becomes

$$q(j) = \sum_i \left[(d+1)p(i) - \frac{1}{d} \right] r(j | i).$$

This is an exact probability-language representation of quantum mechanics, assuming the Hilbert-space/SIC structure. QBism interprets the Born rule as an empirical normative relation supplementing ordinary Bayesian coherence. It therefore shows how Hilbert geometry can be encoded in relations among an agent’s probabilities, but it does **not** show that ordinary Bayesian rationality alone forces those coefficients or the quantum state space. ¹⁰

The contrast with Algorithmic Idealism is especially revealing. Let ordinary quantum mechanics supply a computable experimental measure

$$\mu_{\text{QM}}(o_{n+1} | h_n),$$

where h_n denotes the accumulated experimental record. If Solomonoff induction applies, then the convergence result gives schematically

$$M(o_{n+1} | h_n) \longrightarrow \mu_{\text{QM}}(o_{n+1} | h_n).$$

Thus Algorithmic Idealism can be **asymptotically consistent with quantum mechanics** if quantum theory generates the computable regularities in the observer's history. Müller explicitly uses this kind of argument to establish agreement with ordinary empirical physics under an appropriate physical Church–Turing assumption. ²⁹

But this does not distinguish quantum mechanics from any other simple computable probabilistic law. A classical stochastic universe, a different generalized probabilistic theory, or a computable post-quantum theory could also be learned by universal induction if it generated the data. To genuinely *derive* QM, one needs a result approximately of the form

Self States + Universal Induction + minimal composition assumptions

⇒ non-Kolmogorovian operational structure

⇒ Tsirelson-limited nonlocality + purification + complex Hilbert space

⇒ $P(o) = \text{Tr}(\rho E)$.

No such theorem currently exists in Algorithmic Idealism. Müller says explicitly that reconstructing Bell violation plus no-signalling and BQP computation remains a central goal. ²

Hoffman's program is one level further from this target. The 2014 conscious-agent theory rigorously defines interacting agents and their Markov dynamics and shows that a particular quantum free-particle wavefunction has the same mathematical form as harmonic functions associated with an induced spacetime Markov chain. ²¹ Later work develops networks and fusions of conscious agents beyond spacetime. ³⁰ Such results demonstrate a **mathematical correspondence**, not a derivation of quantum theory. To reach that standard, the theory would need to derive complex amplitude addition, arbitrary finite-dimensional state spaces, tensor products, Bell correlations with the Tsirelson rather than merely no-signalling bound, POVM probabilities, reversible unitary groups, and ultimately relativistic quantum field theory.

Wolfram's program is more structurally quantum-oriented. Multiway evolution naturally provides branching alternatives; branchial graphs encode relations among branches; observer equivalencing is proposed as the origin of familiar quantum phenomenology. Wolfram argues that computational boundedness and persistence make observers perceive quantum mechanics. ³ There are also formal results showing that ZX-calculus rewrites used in categorical quantum information can be embedded into Wolfram-model multiway systems and that relevant multiway/branchial structures possess compatible monoidal structure.

³¹ But an **embedding of an already-quantum calculus into a universal rewriting framework is not a derivation explaining why physical observers must obey exactly that calculus**. The distinction is the same one Müller emphasizes when warning that quantum-looking mathematical objects are insufficient.

²

The resulting scorecard is stark:

Program	State space	Superposition	Entanglement	Unitary/ reversible evolution	Born rule	Overall status
Masanes–Müller reconstruction	Derived	Derived once Hilbert structure obtained	Derived via composition	Derived/ constrained from reversible dynamics	Derived as state–effect pairing	Rigorous, conditional on operational axioms. ¹
CDP purification reconstruction	Derived	Derived	Central via purification/ composition	Derived operationally	Recovered within reconstructed QT	Rigorous, conditional on six informal principles
Hardy reconstruction	Derived	Derived	Recovered in resulting QT	Continuous reversible dynamics central	Trace probability rule recovered	Rigorous under five axioms
QBism	Quantum state space reformulated probabilistically	Present	Present	Present	Exact normative reformulation	Exact interpretation reformulated, not derived from Bayesianism alone. ¹⁰
Algorithmic Idealism	No Hilbert derivation	Open	Bell/ nonclassicality mechanism only sketched	Open	Not derived	Rigorous emergent results; quantum reconstruction explicitly unfinished
Hoffman conscious agents	Markov state spaces, not quantum state space	Wavefunction-like partial correspondence	Not generally derived	Markov evolution, not general unitary QM	Not derived	Partial/ speculative QM reconstruction ⁴
Wolfram/ ruliad	Branchial/ multiway candidate structure	Proposed from branching	Proposed/ structural analogues	Proposed from causal/ multiway properties	No generally accepted complete derivation	Developed conceptual program + partial formal correspondence ³²

Program	State space	Superposition	Entanglement	Unitary/ reversible evolution	Born rule	Overall status
Constructor theory	Generic task/ substrate structure	Only in superinformation theories	Qualitatively recovered	Depends on underlying constructor laws	Probability program does not uniquely reconstruct complex QM	Strong qualitative quantum- information framework unique QM derivation
Spekkens toy theory	Classical ontic + epistemic restriction	Quantum-like analogue	Quantum-like analogues	Restricted classical permutations	Classical probabilities	Deliberately incomplete and KS fails show where simple epistemic accounts fail

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The most significant conceptual lesson from Spekkens is that **recovering interference-looking behavior, no-cloning analogues, teleportation analogues, or even some entanglement-like phenomena is not enough**. His toy theory produces an impressive catalogue of quantum-like effects while still failing exactly at Bell nonlocality and Kochen–Specker contextuality. ¹³ Any Zero Worlds claim of having “derived quantum mechanics” should therefore be required at minimum to derive the genuinely nonclassical correlation structure, not merely a wave equation.

Pusey–Barrett–Rudolph adds a complementary constraint. Under their preparation-independence assumption, models in which distinct quantum wavefunctions merely represent overlapping probability distributions over a deeper underlying physical state conflict with quantum predictions. ¹⁷ This does **not** refute QBism or strong idealism, because those approaches need not posit the kind of underlying physical ontic state assumed in the theorem. Rather, it shows that a naïve “the wavefunction is just ignorance about ordinary hidden matter” strategy is highly constrained.

Empirical support, distinctive predictions, and falsifiability

The empirical record supports **quantum mechanics**, not any one metaphysical explanation of it. A loophole-free Bell experiment by Hensen and collaborators obtained a CHSH value $S = 2.42 \pm 0.20$, rejecting its tested local-realist null at the reported significance level while agreeing with quantum predictions. ⁷ Later and parallel experiments have strengthened Bell tests considerably. What such experiments establish is that candidate foundations must reproduce nonclassical correlations without falling back on the particular conjunction of assumptions excluded by Bell-type theorems. They do not establish that consciousness, observers, computation, or information is ontologically fundamental.

This makes Bell nonlocality an especially valuable **benchmark rather than evidence for Zero Worlds**. A successful observer-first theory should derive not merely

$$S > 2,$$

but the quantum boundary

$$|S| \leq 2\sqrt{2},$$

while excluding stronger Popescu–Rohrlich-type no-signalling correlations. Explaining why the world is quantum rather than merely nonlocal and non-signalling would be a much stronger achievement than deriving qualitative “entanglement.” Müller’s 2026 paper correctly singles out Bell violation together with no-signalling as an unfinished target. ²

Extended Wigner-friend scenarios are more directly relevant to first-person ontology. Bong and collaborators established a “local friendliness” no-go theorem and performed a photonic proof-of-principle experiment whose quantum correlations conflict with a conjunction of assumptions associated with observer-independent outcomes and locality. ⁸ Müller uses this family of results to motivate the idea that a physical theory may not always provide a single global joint probability distribution for every putative observation by every agent. ³⁴ But it would be an overreach to say that such experiments prove “facts are literally created by consciousness.” The theorem constrains an explicit bundle of mathematical assumptions; its experimental “friends” are controlled physical information-processing systems, not demonstrations that phenomenal consciousness is causally required.

Hoffman’s empirical situation is quite different. Fitness-Beats-Truth provides a theorem inside an evolutionary/game-theoretic model showing that fitness optimization need not coincide with truth-tracking perception and, under the paper’s assumptions, can outperform it. ²⁰ This bears on the epistemology of perception. It does not test the ontological claim that reality consists of conscious agents, nor does it constitute evidence for the proposed derivation of particles from their dynamics. The latter needs its own quantitative physical predictions.

Wolfram’s Physics Project has produced mathematical work on discrete rewriting systems, causal graphs and categorical quantum correspondences, but in the primary materials surveyed here I found **no experimentally confirmed parameterized deviation from standard quantum mechanics uniquely predicted by the ruliad/observer theory**. The program presently emphasizes derivational universality rather than a distinct numerical deviation. Wolfram himself states that further details—including specific particle physics and parameters—remain to be derived. ³

Algorithmic Idealism does make claims that differ in principle from ordinary world-based physical theories, but they are concentrated in exotic first-person situations. Müller predicts phenomena such as **probabilistic changelings**, circumstances in which first-person continuation probabilities can diverge from the ordinary third-person physical evolution assigned to the corresponding structure; he also gives nonstandard answers to simulation shutdown, duplication, teletransportation and Boltzmann-brain problems. ³⁵ These are genuine conceptual predictions of the framework rather than mere reinterpretations, but they are not currently accessible as ordinary laboratory tests with publicly comparable outcome frequencies.

Program	Existing empirical support relevant to it	Distinctive prediction beyond standard QM?	Current empirical status
Algorithmic Idealism	Solomonoff convergence is mathematical, while ordinary physics is recovered asymptotically if its probabilities are computable. ²⁶	Yes in principle: simulation/duplication continuity, probabilistic changelings, possible agent-relative behavior in private experiments. ²⁷	Not yet conventionally tested. No complete distinct Born/Bell prediction currently available.
Hoffman	Evolutionary FBT result supports nonveridical-interface possibility. ²⁰	A mature theory should predict particle/scattering/quantum statistics from conscious-agent networks; no such complete distinctive suite yet.	Conscious realism not directly empirically confirmed.
Wolfram	Formal computational correspondences; relativity/QM structures argued from observer theory. ³²	Potentially discreteness/branchial or particle-physics predictions once detailed model is completed.	No distinctive confirmed quantum deviation identified in surveyed primary material.
QBism	Uses exactly standard quantum predictions.	No, as ordinarily formulated; it changes interpretation and normative reading rather than frequencies. ¹⁰	Empirically equivalent to QM.
Operational reconstructions	Recover experimentally successful QM once axioms are adopted.	Generally no different predictions; they explain why the formalism follows from principles. ¹¹	Mathematical explanatory programs, not rival empirical theories.
Constructor theory	Compatible with QM when instantiated quantum mechanically.	Depends on additional constructor-theoretic laws; generic framework alone does not fix new frequencies. ¹⁶	Framework-level, not one competing probability law.

Program	Existing empirical support relevant to it	Distinctive prediction beyond standard QM?	Current empirical status
Spekkens toy theory as full physics	Reproduces many qualitative phenomena.	Predicts absence of specifically quantum Bell/KS phenomena.	Ruled out as full quantum theory by precisely the effects it acknowledges it cannot reproduce. ³⁶
Simple local ψ-epistemic ontological models	Historically motivated hidden-variable picture	Different underlying ontology	Broad subclasses constrained by Bell and PBR-style no-go results. ³⁷

A credible Zero Worlds program should therefore adopt a stronger standard of falsifiability than “it reproduces things quantum mechanics already explains.” At least one of three outcomes is needed.

First, it could **derive standard QM uniquely** from fewer and independently motivated assumptions. That would be explanatory progress even without new empirical frequencies, much as thermodynamic or information-theoretic derivations can explain a pre-existing formalism.

Second, it could generate a **quantitative deviation** in a new regime: for example, a specified modification to Born probabilities in a carefully defined observer-copying experiment,

$$P_{\text{ZW}}(o) - P_{\text{QM}}(o) = \Delta(o) / 0, \quad =$$

with Δ fixed before data collection.

Third, it could supply a **no-go theorem** showing that certain apparently coherent forms of post-quantum physics are impossible given the observer-first postulates. Deriving the Tsirelson bound, information causality, purification, or the impossibility of superquantum no-signalling correlations would count as strong progress even before a new experiment.

Critical assessment: consistency, explanatory power, parsimony, and physicalism

The deepest merit of Zero Worlds thinking is methodological: it reverses the usual explanatory direction. Standard physics ordinarily begins with systems and laws and then asks how observers arise inside them. Zero Worlds asks whether the statistical structure of a lawful external reality can instead be derived from constraints on first-person continuation. Müller’s 2020 result demonstrates that this reversal is not empty philosophy: universal induction can rigorously produce the appearance of stable computable probabilistic environments under appropriate conditions. ³⁸ The 2026 adversarial collaboration between Müller and Kelvin McQueen makes the disagreement particularly clear: physical realism takes a world and laws as fundamental, whereas Algorithmic Idealism makes first-person states fundamental and treats shared worlds as emergent. ³⁹

Its **internal-consistency advantage** is that it does not need to identify a unique global physical state in situations where the very applicability of such a state is contested. This is attractive in Wigner-friend and observer-duplication problems. But the same feature creates a consistency burden: a mature theory must prove when separate first-person histories admit a stable common-world approximation, how communication between observers is reconstructed, and exactly when that approximation fails. Müller has results connecting first- and third-person probabilities in information-rich regimes, while also predicting departures such as changelings when the relevant conditions fail. ³⁵

Algorithmic Idealism is **syntactically parsimonious**—two conceptual postulates in the 2026 formulation—but not obviously computationally or mathematically cheap. Universal induction quantifies over programs for a universal machine and is uncomputable in exact form. Moreover, the invariance theorem only controls changes between universal description languages up to constants; those constants are harmless asymptotically in many induction results but potentially troublesome for exact finite first-person probabilities. The program therefore needs a stronger account of representation invariance if it is to produce precise finite experimental predictions rather than asymptotic regularity claims. Müller explicitly presents Algorithmic Idealism as a framework admitting multiple mathematical realizations, which makes this an acknowledged rather than merely external problem. ²⁹

Hoffman's theory enjoys a different kind of parsimony: it seeks one category of primitive—conscious agents—instead of both matter and consciousness. ²¹ Its weakness is that **phenomenal interpretation outruns the mathematics**. A Markov kernel by itself is not consciousness; attaching the word “experience” to elements of X supplies an ontological interpretation rather than deriving phenomenology from the formal structure. Conversely, if any suitable Markov network can be labeled a conscious-agent network, the mathematical theory risks being too permissive unless additional principles sharply identify which structures correspond to which experiences. The impressive task is not demonstrating that Markov chains can reproduce wave-like functions—they can—but deriving the exact empirical restrictions of quantum theory *because the states are conscious*.

The Fitness-Beats-Truth argument has real explanatory value for Hoffman's **interface thesis**, but only limited leverage on the physics. Evolution can explain why a creature's internal perceptual variables need not be isomorphic to whatever causes its sensory inputs. ²⁰ It cannot, without further theory, explain why the hidden reality behind that interface has the complex Hilbert-space dimension, gauge structure and Hamiltonians of our universe. The move from “perception is not veridical” to “conscious agents are fundamental ontology” is abductive, not deductive.

Wolfram's framework has almost the inverse parsimony profile. It is economical in selecting a universe—there is no arbitrary choice of a single rule at the ultimate ruliad level because all computations are included—but extravagant in ontology: **everything computationally possible is present in the underlying abstract structure**. ⁴⁰ Explanatory selection is shifted from “why this physical rule?” to “why observers like us sample this part/equivalence class of the ruliad in this way?” Wolfram attempts to answer this with computational boundedness and persistence. His own observer-theory discussion acknowledges circularities concerning what makes a persistent observer and what counts as equivalencing the same states, indicating that the observer theory itself still needs sharpening. ²²

The formal embedding of ZX-calculus into multiway rewriting is valuable because it proves a real structural compatibility between a universal rewrite framework and categorical quantum processes. ³¹ Yet precisely because universal rewriting systems can embed an enormous range of computational structures,

embeddability has weak selecting power. To explain physics, one needs inevitability: why bounded persistent observers must get complex quantum theory and cannot get some other internally coherent probabilistic calculus.

QBism is conceptually more conservative. It offers a disciplined first-person interpretation of probabilities without denying the existence of external systems. The Born rule becomes a constraint relating the agent’s probability assignments rather than the manifestation of a physically real wavefunction. ¹⁰ Its virtue is that it stays exactly aligned with actual quantum practice. Its limitation for Zero Worlds purposes is equally clear: the central quantum structure remains an empirically supplied normative constraint. It relocates the mystery more than it eliminates it.

The operational reconstructions are almost the mirror image. They have the **strongest mathematical derivations and weakest metaphysical ambitions**. If purification, local tomography, continuous reversibility and related information principles hold, the theorems show that quantum structure follows. ⁵ But a critic may reasonably ask whether principles such as purification already encode much of what is distinctly quantum. The response is that this is still explanatory compression: six operational statements are conceptually more transparent than simply postulating arbitrary complex Hilbert spaces, tensor products and trace rules. Yet a Zero Worlds theory could go further by explaining *why those operational principles themselves emerge*.

Constructor theory is useful here as a foil. It changes the explanatory language from trajectories to possible and impossible tasks and succeeds in identifying a common structural source for such quantum-information phenomena as no-cloning, incompatibility, unpredictable measurement and entanglement in superinformation media. ³³ But Deutsch and Marletto explicitly insist that information is physical rather than a free-floating abstraction. Constructor theory is therefore a potential formal partner for observer-first programs but not evidence for idealism. ¹⁶

Compared with all these approaches, **standard physicalist quantum theories have a decisive empirical-development advantage**. They already possess quantitative Hamiltonians, quantum field theory, scattering amplitudes, symmetry principles, particle spectra and precision predictions. The foundational cost is that they begin much further downstream: quantum fields, wavefunctions, particles, spacetime or dynamical laws are taken as physically real or otherwise primitive. Observer-first programs aim to explain why this effective physical structure exists at all, but have not yet matched its quantitative reach.

Criterion	Zero Worlds / Algorithmic Idealism	Hoffman	Wolfram	Operational reconstructions	Standard physicalist starting point
Ontological economy	Very high: no primitive external world	High if consciousness alone suffices	Mixed: one ruliad, but contains all computation	Neutral	Lower at primitive level: physical state/law assumed

Criterion	Zero Worlds / Algorithmic Idealism	Hoffman	Wolfram	Operational reconstructions	Standard physicalist starting point
Formal economy	Small postulate set, but universal- induction machinery is powerful and uncomputable	Standard Markov machinery; phenomenological interpretation extra	Very large universal rewrite structure	Compact axioms + mature convex mathematics	Mature but structurally rich formalism
Derivation of ordinary objective world	Genuine asymptotic results. ³⁸	Programmatic emergence from agent networks	Observer coarse-graining argument	World/system framework operationally assumed	World primitive
Derivation of full finite QM	No	No	Not yet in reconstruction- theorem sense	Yes, under axioms	QM assumed or independently motivated
Born rule from deepest primitives	Open	Open	Incomplete	Recovered after operational axioms	Usually postulated/ theorem inside chosen formalism
QFT / Standard Model	Not derived	Not derived	Research target	Most reconstructions stop at finite- dimensional QM	Fully developed
Unique testable predictions	Exotic observer scenarios; not yet standard- lab discriminators	No mature distinctive quantum- frequency predictions	No mature confirmed distinctive quantum- frequency prediction found	Usually none; reproduces QM	Some physicalist modifications, such as collapse models, can be directly empirically different
Best explanatory target	Why there appears to be any world/law at all	Why spacetime perception is an interface to consciousness	Why physical laws emerge from computation + observer constraints	Why quantum kinematics has its peculiar information- theoretic form	What physical systems quantitatively do

The crucial parsimony lesson is therefore:

$$\boxed{\text{few primitive nouns} /}$$

= few explanatory

A theory with “only consciousness,” “only self-states,” or “only the ruliad” wins ontological economy only if its mathematical laws can recover the enormous specific structure currently placed in the physical ontology. Otherwise complexity has merely moved from the noun list into the transition rule, observer measure, or interpretation map.

Open problems and a prioritized research agenda

The best next step is not another qualitative claim that “quantum mechanics resembles observer dependence.” It is to turn Zero Worlds into a **reconstruction problem with explicit target theorems and failure conditions**.

Priority one: prove an observer-to-GPT bridge theorem. Begin from a clearly specified first-person theory—Müller’s self-state model is currently the cleanest candidate—and define interventions, repeatable experiments and operational equivalence solely in its native language. The target is to derive a convex operational state space

$$\Omega, \quad e : \Omega \rightarrow [0, 1], \quad T : \Omega_A \rightarrow \Omega_B$$

without assuming a physical system underneath. The central theorem should state conditions under which persistent regularities in self-states induce well-defined preparation, transformation, measurement and composition equivalence classes. This would connect Algorithmic Idealism to the mature GPT reconstruction literature rather than trying to reinvent quantum formalism directly. Müller already recognizes quantum reconstructions as the relevant benchmark. ⁴¹

Priority two: derive nonclassical composition rather than merely probability. Universal induction is naturally Kolmogorovian at the meta-level, so the difficult issue is explaining why operational observables do not all fit into a single ordinary joint probability distribution. The target should be a theorem showing that first-person restrictions, forgetting, or incompatible counterfactual continuations generate contextual operational probability spaces. Müller explicitly suggests this possibility and points toward Bell/no-signalling, but the mechanism remains unfinished. ⁴²

A decisive benchmark would be

$$\text{observer-first postulates} \implies 2 < S_{\max} = 2\sqrt{2} < 4,$$

not merely $S > 2$. The appearance of the **Tsirelson bound** would strongly discriminate a specifically quantum reconstruction from a generic contextual or no-signalling one.

Priority three: derive one existing reconstruction axiom set. Rather than deriving Hilbert space all at once, choose one proven theorem as the destination. CDP is especially attractive because **purification** provides a clean nonclassical principle; Masanes–Müller may be particularly natural because Markus Müller is already central to both sides of the research bridge. ¹¹

For example, seek proofs of:

Persistent intersubjective agreement	$\Rightarrow$ causality,
observer-state compositionality	$\Rightarrow$ local tomography,
reversible informational equivalence	$\Rightarrow$ pure-state transitivity,
loss of inaccessible information / first-person completion	$\Rightarrow$ purification?

The question mark matters. Purification should not be smuggled in by metaphor; it needs a genuine theorem.

Priority four: obtain the Born rule without assuming the quantum measure. Algorithmic Idealism currently learns any sufficiently simple computable measure. The key measure-selection problem is therefore:

Why does universal induction favor quantum probability structure, rather than merely learn it after observing it?

A successful answer might involve invariance under changes of self-state encoding, composition of independent experiments, symmetry of informational alternatives, or a decision/coherence principle. QBism's SIC formulation supplies a useful target equation because it expresses the quantum modification to ordinary total-probability reasoning transparently. ¹⁰

One especially clean research goal would be to derive coefficients equivalent to

$$(d+1)p(i) - \frac{1}{d}$$

rather than insert them. If those coefficients emerged from an observer-first invariance principle, the connection between Zero Worlds and QBism would become mathematically substantive rather than analogical.

Priority five: derive continuous dynamics and locality. Even a successful finite-dimensional reconstruction is only the beginning. One must explain why reversible time evolution forms the appropriate continuous unitary group and why observed interactions are generated by approximately local Hamiltonians:

$$U(t) = e^{-iHt/\hbar}.$$

Then come Lorentz symmetry, fields, gauge structure, fermionic statistics, renormalization and the Standard Model. Operational reconstruction research proves that quantum kinematics can be highly constrained by information principles, but it does not explain the actual Hamiltonian content of our universe. ¹¹

Wolfram's discrete causal/rewrite machinery may ultimately be more naturally suited than Algorithmic Idealism to the spacetime/locality side of this problem. A serious synthesis would therefore not simply identify the ruliad with Solomonoff induction. It would ask whether **algorithmic weighting can supply a measure over observer-accessible sectors of a multiway/rulial structure**, while the multiway causal framework supplies compositional and geometric constraints. At present this is a proposed research

synthesis, not an established equivalence between Müller and Wolfram. Their fundamental ontologies are different. ⁴³

Priority six: formalize memory, forgetting and identity. Müller’s simplest bit model accumulates information monotonically, whereas real observers forget, compress and rewrite memories. His 2026 discussion explicitly treats alternative formalizations and notes the importance of models beyond the preliminary bit construction. ²⁹ This issue is not peripheral: contextuality and Wigner-friend phenomena may depend precisely on the impossibility of putting all possible observer records into one persistent global Boolean history. A forgetting model should specify a stochastic/computable map

$$F : x_t \mapsto \tilde{x}_t$$

and then prove that first-person continuation probabilities remain invariant under irrelevant representational losses while allowing nonclassical operational structure.

Priority seven: pre-register a genuinely discriminating experiment or thought-experiment protocol.

Müller’s strongest distinct predictions concern duplication, simulation and private observer scenarios. ³⁵

To make these scientific rather than merely metaphysical, the program needs a protocol with precisely defined physical operations, clearly identified observer-state variables, competing quantitative predictions, and a criterion for how private outcomes become statistically assessable without silently reintroducing a third-person global world.

A mature test should look like

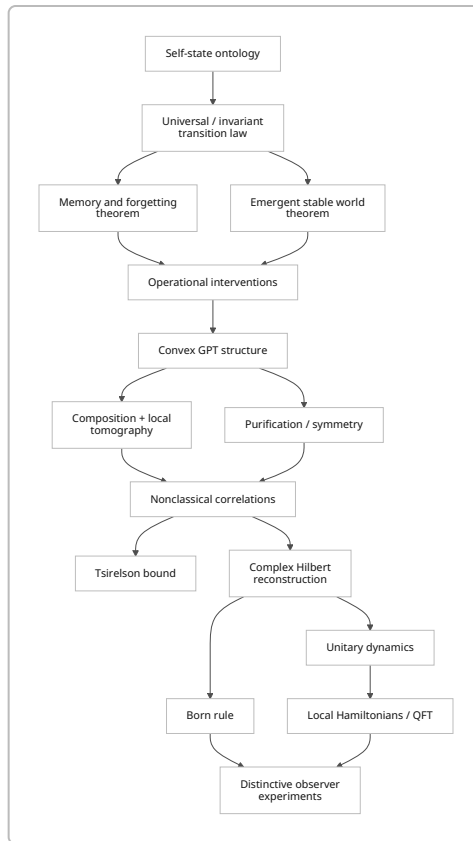
$$H_{QM} : P(o) = p,$$

versus

$$H_{ZW} : P(o) = p + \Delta,$$

with Δ fixed by the theory before the experiment. Until such a calculation exists, “different predictions in exotic cases” should be described as **theoretical discriminators**, not empirical confirmations.

The overall program can be summarized as a dependency graph:



The research strategy should resist shortcuts. A wave equation appearing somewhere in an emergent model is not enough. A complex number is not enough. Branching is not enough. Interference-like behavior is not enough. Spekkens’ toy model demonstrates how many superficially quantum phenomena can arise in a theory that nevertheless fails the decisive nonclassical tests. ¹³ Müller’s own 2026 methodological warning is exactly right: the target is the **experimental statistical structure that is genuinely quantum**.

2

Publication-ready conclusions and primary-source map

A defensible claims-site thesis can be stated at three levels of confidence.

Established: Quantum mechanics can be reconstructed from surprisingly abstract operational and information-theoretic principles without taking Hilbert space, complex amplitudes and the Born trace rule as unexplained primitive axioms. Hardy, Chiribella–D’Ariano–Perinotti, and Masanes–Müller provide rigorous examples. ⁵

Established within a narrower formal model: Müller’s algorithmic first-person framework can make a simple computable probabilistic external world emerge approximately and asymptotically rather than posit it fundamentally. It supplies a mathematically nontrivial example of “law without fundamental world-law.”

⁴⁴ His 2026 formulation extends the conceptual program to self-states, simulation, observer duplication and Wigner-friend-type situations. ³⁴

Not established: that Algorithmic Idealism, conscious-agent ontology, or the ruliad uniquely implies complex Hilbert-space quantum mechanics; that consciousness is required for quantum measurement; that the Born rule has been derived from consciousness; that Bell or Wigner-friend experiments empirically prove idealism; or that any of these non-material ontologies currently predicts ordinary laboratory data better than standard quantum theory. Müller explicitly treats the genuinely quantum reconstruction as unfinished. ² Hoffman’s published conscious-agent work gives partial mathematical correspondences, while Wolfram’s writings advance a broader derivational claim whose detailed completion remains a research program. ⁴⁵

This yields a useful final distinction:

“Matter is not fundamental”
is a metaphysical hypothesis;

“Operational quantum theory follows from principles *A, B, C*”
can be a mathematical theorem;

and

“Those principles themselves follow from first-person existence”
is the central open Zero Worlds problem.

That last proposition is where the scientifically strongest version of this program should concentrate. A successful proof would do something conceptually remarkable: it would transform observer-first metaphysics into a derivational foundation for physics rather than an interpretation appended to physics after the fact.

The following primary-source map is suitable as the basis for a references page or linked bibliography on a claims site.

Area	Primary sources and direct links	Why it matters
Algorithmic Idealism / Zero Worlds	Markus P. Müller, Law without Law: From Observer States to Physics via Algorithmic Information Theory , <i>Quantum</i> 4, 301 (2020); arXiv: 1712.01826 . ⁴⁶	Foundational technical paper: universal induction determines first-person continuation; objective world emerges statistically.
	Markus P. Müller, Algorithmic Idealism: What Should You Believe to Experience Next? , <i>Foundations of Physics</i> 56, 11 (2026). ³⁴	Current conceptual formulation: Self States + State Change; explicitly identifies quantum reconstruction as unfinished.
	Kelvin J. McQueen & Markus P. Mueller, Realism about the External World: An Adversarial Collaboration (2026). ³⁹	Direct 2026 confrontation between physical realism and Algorithmic Idealism.

Area	Primary sources and direct links	Why it matters
Rigorous QM reconstruction	Lluís Masanes & Markus P. Müller, A Derivation of Quantum Theory from Physical Requirements , <i>New Journal of Physics</i> 13, 063001 (2011). ¹⁸	Particularly important bridge: the same Müller participates in a rigorous operational derivation of QM.
	Giulio Chiribella, Giacomo Mauro D'Ariano & Paolo Perinotti, Informational Derivation of Quantum Theory , <i>Physical Review A</i> 84, 012311 (2011). ²³	Derives QT from five information axioms plus purification.
	Lucien Hardy, Quantum Theory From Five Reasonable Axioms (2001). ²⁴	Early influential reconstruction; continuous reversible transformations distinguish quantum from classical structure.
QBism / agent-centered QM	Christopher A. Fuchs & Rüdiger Schack, Quantum-Bayesian Coherence: The No-Nonsense Version , <i>Reviews of Modern Physics</i> 85, 1693 (2013). ¹⁰	Born rule as empirical normative relation among an agent's probabilities; key comparison with first-person approaches.
Hoffman / consciousness-first	Donald D. Hoffman & Chetan Prakash, Objects of Consciousness , <i>Frontiers in Psychology</i> 5, 577 (2014). ²¹	Formal definition of conscious agents and proposed emergence of spacetime/quantum objects.
	Donald Hoffman, Chetan Prakash & Robert Prentner, Fusions of Consciousness , <i>Entropy</i> 25, 129 (2023). ³⁰	Develops consciousness-first dynamics beyond spacetime.
	Chetan Prakash et al., Fitness Beats Truth in the Evolution of Perception , <i>Acta Biotheoretica</i> 69, 319–341 (2021). ²⁰	Formal evolutionary support for perception-as-interface; should not be conflated with proof of conscious realism.
Wolfram / computational universe	Stephen Wolfram, Observer Theory (2023). ²²	Most direct account of computational boundedness, persistence and observer-dependent emergence of physical law.
	Stephen Wolfram, What Ultimately Is There? Metaphysics and the Ruliad (2026). ⁴⁰	Current statement of the ruliad ontology and claimed emergence of GR, QM and thermodynamics.
	Jonathan Gorard, Manojna Namuduri & Xerxes D. Arsiwalla, ZX-Calculus and Extended Hypergraph Rewriting Systems I (2020). ³¹	Formal embedding of categorical quantum ZX calculus in multiway rewriting; important but not itself a derivation of QM.

Area	Primary sources and direct links	Why it matters
Constructor theory	David Deutsch, Constructor Theory (2012). ⁴⁷	Recasts fundamental laws as possible/impossible transformations.
	David Deutsch & Chiara Marletto, Constructor Theory of Information (2014/2015). ³³	Derives a family of quantum-information phenomena from “superinformation” structure while explicitly keeping information physical.
Ontic / epistemic models	Robert W. Spekkens, In Defense of the Epistemic View of Quantum States: A Toy Theory (2004/2007). ¹³	Essential warning that many quantum-looking effects can arise without full quantum nonclassicality.
	Matthew Pusey, Jonathan Barrett & Terry Rudolph, On the Reality of the Quantum State , <i>Nature Physics</i> 8, 475–478 (2012). ¹⁷	Constrains overlapping ψ -epistemic hidden-state models under preparation independence.
Empirical benchmarks	Bas Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres , <i>Nature</i> 526, 682–686 (2015); arXiv: 1508.05949 . ⁴⁸	Minimum benchmark for any proposed emergence of entanglement/nonlocal quantum statistics.
	Kok-Wei Bong et al., A Strong No-Go Theorem on the Wigner’s Friend Paradox , <i>Nature Physics</i> 16, 1199–1205 (2020). ⁸	Directly relevant benchmark for theories that make observer-relative facts fundamental.
Historical “Johannes Müller” candidate	Johannes Peter Müller, <i>Handbuch der Physiologie des Menschen</i> and the doctrine of specific nerve energies; modern historical discussion summarized by Lohff. ¹⁹	Important precursor to interface-dependent perception, but not the Müller behind Zero Worlds or a quantum-foundations reconstruction.

The most productive synthesis is thus **not** “Hoffman + Wolfram + Müller all say consciousness creates quantum physics.” That would erase their substantive disagreements and overstate their results. The stronger research thesis is that they explore three distinct ways of moving the physical world away from the bottom level—**first-person algorithmic states, phenomenal conscious agents, and universal computation**—while quantum-reconstruction theory supplies a separate body of rigorous mathematics describing the intermediate principles sufficient to recover quantum structure. ⁴⁹

The outstanding scientific problem is to close that gap:

non-material primitive $\implies$ observer/information principles $\implies$ quantum operational geometry $\implies$ Born rule
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At present, the middle implication is mathematically well developed **once the operational principles are assumed**; the first implication is where Zero Worlds remains genuinely frontier research, and the final

implication to realistic quantum field theory remains largely open even for the best information-theoretic reconstructions. ⁶

That is both the principal limitation and the principal opportunity: **a Zero Worlds program would become a major foundations result not by finding another analogy between consciousness and wavefunctions, but by proving why first-person existence forces the precise probability geometry of quantum theory.**

¹ ¹⁴ ²⁵ ³⁸ ⁴⁴ ⁴⁶ Law without law: from observer states to physics via algorithmic information theory – Quantum

https://quantum-journal.org/papers/q-2020-07-20-301/?utm_source=chatgpt.com

² ⁹ ²⁶ ²⁷ ²⁹ ³⁵ ⁴¹ Algorithmic Idealism: What Should You Believe to Experience Next? | Foundations of Physics | Springer Nature Link

<https://link.springer.com/article/10.1007/s10701-026-00913-1>

³ ¹⁵ ²² ³² ⁴³ Observer Theory—Stephen Wolfram Writings

https://writings.stephenwolfram.com/2023/12/observer-theory/?utm_source=chatgpt.com

⁴ ²¹ ⁴⁵ Frontiers | Objects of consciousness

https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2014.00577/full?utm_source=chatgpt.com

⁵ ⁶ ¹¹ ¹⁸ ²⁸ A derivation of quantum theory from physical requirements

https://arxiv.org/abs/1004.1483?utm_source=chatgpt.com

⁷ ⁴⁸ Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres | Nature

https://www.nature.com/articles/nature15759?utm_source=chatgpt.com

⁸ A strong no-go theorem on the Wigner’s friend paradox | Nature Physics

https://www.nature.com/articles/s41567-020-0990-x?utm_source=chatgpt.com

¹⁰ Quantum-Bayesian Coherence: The No-Nonsense Version

https://arxiv.org/abs/1301.3274?utm_source=chatgpt.com

¹² ⁴⁷ Constructor Theory

https://arxiv.org/abs/1210.7439?utm_source=chatgpt.com

¹³ ³⁶ In defense of the epistemic view of quantum states: a toy theory

https://arxiv.org/abs/quant-ph/0401052?utm_source=chatgpt.com

¹⁶ ³³ Constructor Theory of Information

https://arxiv.org/abs/1405.5563?utm_source=chatgpt.com

¹⁷ ³⁷ On the reality of the quantum state

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https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470015902.a0002428?utm_source=chatgpt.com

²⁰ Fitness Beats Truth in the Evolution of Perception - Rutgers, The State University of New Jersey

https://www.researchwithrutgers.org/en/publications/fitness-beats-truth-in-the-evolution-of-perception/?utm_source=chatgpt.com

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24 Quantum Theory From Five Reasonable Axioms

https://arxiv.org/abs/quant-ph/0101012?utm_source=chatgpt.com

30 Fusions of Consciousness

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https://link.springer.com/article/10.1007/s10701-026-00913-1?utm_source=chatgpt.com

39 Realism about the external world: an adversarial collaboration

https://arxiv.org/abs/2607.16379?utm_source=chatgpt.com

40 What Ultimately Is There? Metaphysics and the Ruliad—Stephen Wolfram Writings

https://writings.stephenwolfram.com/2026/02/what-ultimately-is-there-metaphysics-and-the-ruliad/?utm_source=chatgpt.com